

Euclidean Geometry In Mathematical Olympiads

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Lemma 1.48 (Simson Line)



fgm

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[Solution \(no phantom no directed\)](#)

👁 This is a simple angle chase problem. Let $\angle PYZ = x$. We see $\angle PAZ = x$ by cyclic $PAZY$ and consequentially we know $\angle PAB = 180 - x$. Then by cyclic $ABCP$ we know $\angle PCB = x$ and $\angle PYX = 180 - x$. Therefore $\angle PYZ + \angle PYX = 180$ and we are done.

[Solution no phantom but directed](#)

👁 We know $\angle PYZ = \angle PAZ = \angle PAB = \angle PCB = \angle PYX$ so ZYX are collinear.

[phantom points no directed](#)

👁 Let ZY intersect BC at X' . Our goal is to show $PX \perp BC$. We can do this by showing either $PYXC$ is cyclic or $PZAY$ is cyclic. Let $\angle PYZ = x$. Now we see $\angle ZAP = x$ and $\angle PAB = 180 - x$ and so $\angle PCB = x$. Now $\angle PYX = 180 - x$ so $PYXC$ is cyclic and we are done.

[phantom and directed](#)

👁 We know $\angle PYZ = \angle PAZ = \angle PAB = \angle PCB = \angle PYX$ so we know $PXCY$ is cyclic and $\angle PYC = \angle PXC = 90^\circ$ and we are done.

programming_...
1 post

Oct 17, 2020, 5:47 PM

#7

Is it necessary to use directed angles here? I solved it without them.

DepressedCubic
15 posts

Oct 29, 2020, 12:21 AM • 1 👍

#8

“ programming_wiz wrote:

Is it necessary to use directed angles here? I solved it without them.

Notice that there are actually many possible configurations for the problem depending on